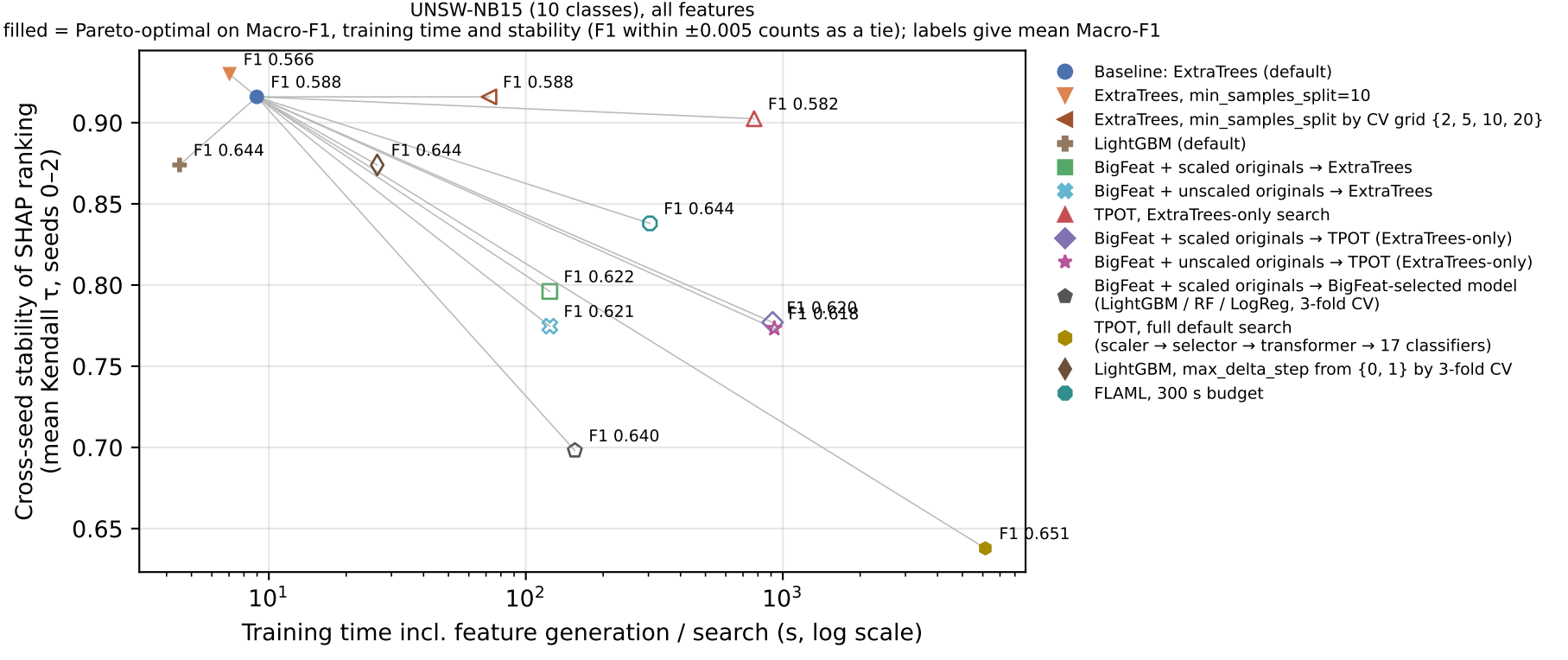
UNSW-NB15 (10 classes), all features
filled = Pareto-optimal on Macro-F1, training time and stability (F1 within ±0.005 counts as a tie); labels give mean Macro-F1
Cross-seed stability of SHAP ranking
(mean Kendall τ, seeds 0–2)
0.90
0.85
0.80
0.75
0.70
0.65
10^1
10^2
10^3
Training time incl. feature generation / search (s, log scale)
F1 0.566
F1 0.588
F1 0.588
F1 0.582
F1 0.644
F1 0.644
F1 0.644
F1 0.622
F1 0.621
F1 0.620
F1 0.618
F1 0.640
F1 0.651
Baseline: ExtraTrees (default)
ExtraTrees, min_samples_split=10
ExtraTrees, min_samples_split by CV grid {2, 5, 10, 20}
LightGBM (default)
BigFeat + scaled originals → ExtraTrees
BigFeat + unscaled originals → ExtraTrees
TPOT, ExtraTrees-only search
BigFeat + scaled originals → TPOT (ExtraTrees-only)
BigFeat + unscaled originals → TPOT (ExtraTrees-only)
BigFeat + scaled originals → BigFeat-selected model
(LightGBM / RF / LogReg, 3-fold CV)
TPOT, full default search
(scaler → selector → transformer → 17 classifiers)
LightGBM, max_delta_step from {0, 1} by 3-fold CV
FLAML, 300 s budget